

Math - 4

statements:

1. Complex wave: A complex wave is formed by superposition (addition) of multiple simple sinusoidal waves (each with its own frequency, amplitude & phase).

$x(t) = A_1 \sin(2\pi f_1 t) + A_2 \sin(2\pi f_2 t + \phi)$

Amplitude frequency phase

2. speech & fourier series: speech signals are quasi-periodic and can be expressed using fourier series, which decomposes signals into sums of sinusoidal components.

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{j n \omega_0 t}, \text{ where } \omega_0 = \frac{2\pi}{T}$$

3. Trigonometric form of Fourier series:

A periodic function can be expressed as a sum of sine & cosine terms.

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

where,

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt$$

4. Complex form of Fourier series:

Uses exponential functions to represent periodic signals more compactly.

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

5. stable signal: A signal is stable if its total energy is finite, eqn:

continuous $\rightarrow$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$$

discrete $\rightarrow$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

6. Unstable signal: A signal is unstable if its energy tends to infinity or grows unbounded.

$$x(t) = e^t, t \geq 0 \Rightarrow \text{unstable}$$

7. periodic signal: A signal that repeats after a time period T .

$$x(t) = x(t+T)$$

example $\rightarrow$

$$x(t) = \cos(2\pi t), T=1$$

8. Aperiodic signal: A signal that doesn't repeat for any finite period.

$$x(t) = e^{-t} u(t)$$

9. Line spectrum: Graphical representation of frequency components of a signal.

→ Discrete line spectrum: for periodic signal.

→ Continuous ~~line~~ spectrum: for aperiodic signal.

10. Convolution sum (Discrete-Time):

It is a mathematical operation that describes the output of a linear time-invariant (LTI) discrete-time system in response to any arbitrary input signal.

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

12. Convolution Integral (Continuous time):

It is a fundamental operation used to determine the output of a continuous-time linear-time-invariant (LTI) system given its input and impulse response.

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t - \tau) d\tau$$

13. Unit step function: It is a piecewise-defined function that represents a signal which turns on at $t = 0$ and remains on indefinitely.

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

14. Ramp function: It is a linearly increasing function that starts at zero for $t = 0$ and increases proportionally with time.

$$r(t) = t \cdot u(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases}$$

15. Impulse function:

It is a generalized function that represents an instantaneous spike at $t = 0$, having infinite amplitude and zero width, but an area of one under its ~~curve~~ curve.

$$\delta(t) = 0 \text{ for } t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$x(t) * \delta(t) = x(t)$$

16. Continuous vs Discrete:

→ A discrete spectrum consists of isolated eigenvalues, occurring for compact or finite rank operators. A continuous spectrum contains a range of non-isolated values, appearing in differential operators. The spectrum is purely discrete for finite-dimensional operators, while infinite dimensional operators may have both.

Feature	Discrete Spectrum	Continuous Spectrum
Signal Type	Periodic	Aperiodic
Mathematical Tool	Fourier Series	Fourier Transform
Frequency Components	Specific	Infinite, continuous range
Graph Type	Line, Stem plot	Smooth curve
Amplitude Representation	Vertical lines	Continuous amplitude distribution
Duration of signal	Infinite in time	Finite in time